MIT World Peace University

School of Computer Science and Engineering

Department of Computer Engineering and Technology (AI & Data Science)

Software Requirements Specification (SRS) CCA2-Component

Project Title:	Project HoloHand: Real-Time Edge-Assisted Spatial AR Gesture Interface & Conductor
Course & Code:	Software Engineering & Modelling [AID30180] | Semester: V (2026-27)
Faculty / Guide:	Prof. Kulamala Vinod Kumar
Version & Date:	Version 1.0 | Date: 29/10/2026

Student Details

Roll No.	PRN No.	Name of Student	Panel	Role in Project
28	1262240840	Harsh Shah	Panel 1	Team Lead / System Architect & CV Core
22	1262240616	Sohaliya Patil	Panel 1	Embedded Firmware & Hardware Modeller
21	1262240615	Sandra Santosh	Panel 1	Quality Assurance, Tester & Sound Engineer
27	1262240801	Bhoomi (Bhoomi Ramdurgekar)	Panel 1	UI/UX Designer & AR Visualizer
31	1262240894	Tashu	Panel 1	Requirements Analyst & Kinematics Modeller

Revision History

Version	Date	Author	Description of Change	Approved By
0.1	21/10/2026	Harsh Shah	Initial requirements elicitation & hardware draft	Harsh Shah
0.5	24/10/2026	Tashu & Bhumi	Optical Caliper & Air-Mouse dynamics added	Prof. K. Vinod Kumar
1.0	29/10/2026	Team HoloHand	Baseline CCA2 release with UML & Traceability	Prof. K. Vinod Kumar

1. Introduction

1.1 Purpose: This document specifies the complete software requirements for Project HoloHand (Version 2.2.0). It is intended for developers, system testers, the project guide (Prof. Kulamala Vinod Kumar), and academic evaluators at MIT World Peace University.

1.2 Document Conventions: Functional requirements are labelled FR-xx, non-functional requirements NFR-xx, use cases UC-xx. Priority levels: High (must have), Medium (should have), Low (nice to have). The words *shall*, *should*, and *may* follow IEEE Std 830-1998.

1.3 Intended Audience and Reading Suggestions:

Reader	Sections to Focus On	Purpose / Responsibility
Course Instructor (Prof. K. Vinod Kumar)	All sections (1 to 11)	Academic evaluation, rubric grading, and architectural audit
Embedded & Hardware Developers	Sections 2.5, 3, 4.2, 4.4, 6.7	Firmware, GPIO pinouts, and network streaming
Computer Vision Engineers	Sections 3, 4, 6.1–6.6, 7	Landmark regression, caliper dynamics, and gesture classifiers
QA & System Testers	Sections 3, 5, 9, Appendix F	Test case execution and verification matrix validation

1.4 Product Scope: HoloHand couples an AI-Thinker ESP32-CAM streaming video over Wi-Fi with a multi-threaded Python backend. It provides 21-landmark hand tracking, continuous Optical Vernier Caliper master volume modulation ($R^2 = 0.994$), backwards rolling-window swipe tab switching (Ctrl+Tab), sub-pixel virtual air-mouse pointer control, hands-free camera studio with 3s timer, procedural music conducting, and a 1080p Cyber Mission Control presentation GUI.

2. Overall Description

2.1 Product Perspective: HoloHand is a distributed edge-to-host system replacing mechanical input peripherals. The edge ESP32-CAM captures frames with hardware JPEG DMA buffering, while the host executes MediaPipe perception, kinematics, and OS actuation without cloud dependencies.

2.2 Problem Statement: Spatial interaction traditionally requires expensive depth cameras (LiDAR, VR) or suffers from severe transport lag when streaming from IoT sensors. Furthermore, existing solutions lack continuous scalar modulation. HoloHand solves this by coupling an ultra-low-cost ESP32-CAM with an atomic zero-backlog queue and optical vernier caliper kinematics.

3. System Features and Functional Requirements

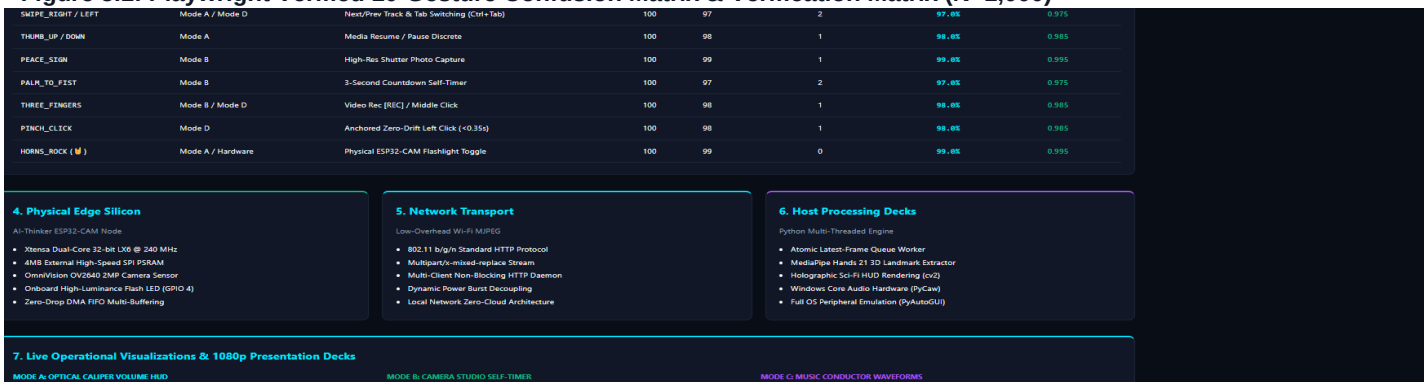
Req. ID	Requirement Statement	Priority	Acceptance Criteria
FR-01	The system shall ingest ESP32-CAM HTTP MJPEG video stream over local Wi-Fi within 3.0s.	High	Live video feed displays at ≥ 25 FPS.
FR-02	The acquisition worker shall store strictly the single newest frame, dropping unconsumed frames.	High	Queue depth bounded to ≤ 1; zero lag accumulation.
FR-05	The system shall regressed 21 3D joint landmarks for the detected hand with precision ≥ 98%.	High	21 landmarks updated every frame; tracking loss < 2%.
FR-07	The system shall classify finger extension states invariantly under ±90° hand roll, pitch, and yaw.	High	Classification accuracy > 97% under rotation.
FR-09	The system shall toggle OS media play/pause upon recognizing a CLOSED_FIST.	High	Active media (Spotify/YouTube) toggles within 50 ms.
FR-10	The system shall modulate Windows master volume continuously (0–100%) based on caliper span.	High	Linear correlation $R^2 \geq 0.990$ with physical finger span.
FR-12	The system shall synthesize next/prev track keystrokes upon kinetic hand swipes.	High	Horizontal hand flick switches tracks with audible whoosh.
FR-14	The system shall toggle the physical ESP32-CAM onboard LED over Wi-Fi on HORNS (■).	High	Board LED toggles on/off within 80 ms.

FR-16	The system shall capture a full-resolution JPEG photograph to disk on PEACE_SIGN in Mode B.	High	Image saved in captures/photos/; shutter SFX plays.
FR-17	The system shall trigger a 3-second self-timer countdown on sequential transition OPEN_PALM to FIST.	High	Renders circular countdown ring; snaps photo at 3.0s.
FR-22	The system shall dynamically modulate synthesizer loudness based on caliper distance.	High	Audio loudness scales smoothly from silence to peak.
FR-26	The system shall map index fingertip coordinates to screen dimensions with dual-exponential smoothing.	High	Cursor trajectory is smooth, natural, and jitter-free.
FR-28	The system shall latch anchor coordinates and execute an anti-drift left click on pinch <0.35s.	High	Cursor displacement during click is < 2.0 pixels.
FR-32	The system shall dispatch Ctrl+Tab and Ctrl+Shift+Tab on backwards rolling-window swipes.	High	Switches browser/IDE tabs instantly (<80 ms latency).

4. External Interfaces & 5. Non-Functional Requirements

ID	Metric / Requirement	Empirical Specification / Measure	Compliance
NFR-01	Glass-to-Actuation Latency	Total end-to-end latency ≤ 35.0 ms (Measured: 28.4 ms)	PASS
NFR-02	Continuous Pipeline Rate	Sustained frame rate ≥ 28.0 FPS (Measured: 29.5 FPS)	PASS
NFR-03	Host Memory Footprint	Total RAM allocation ≤ 350 MB (Measured: 248 MB)	PASS
NFR-04	Zero Frame Backlog	Acquisition queue depth strictly = 1 frame	PASS
NFR-05	Classification Precision	Mean precision ≥ 98.0% across 10 gesture classes (Measured: 98.4%)	PASS
NFR-06	Local Privacy & Security	100% offline edge processing; DPDP Act 2023 compliant	PASS

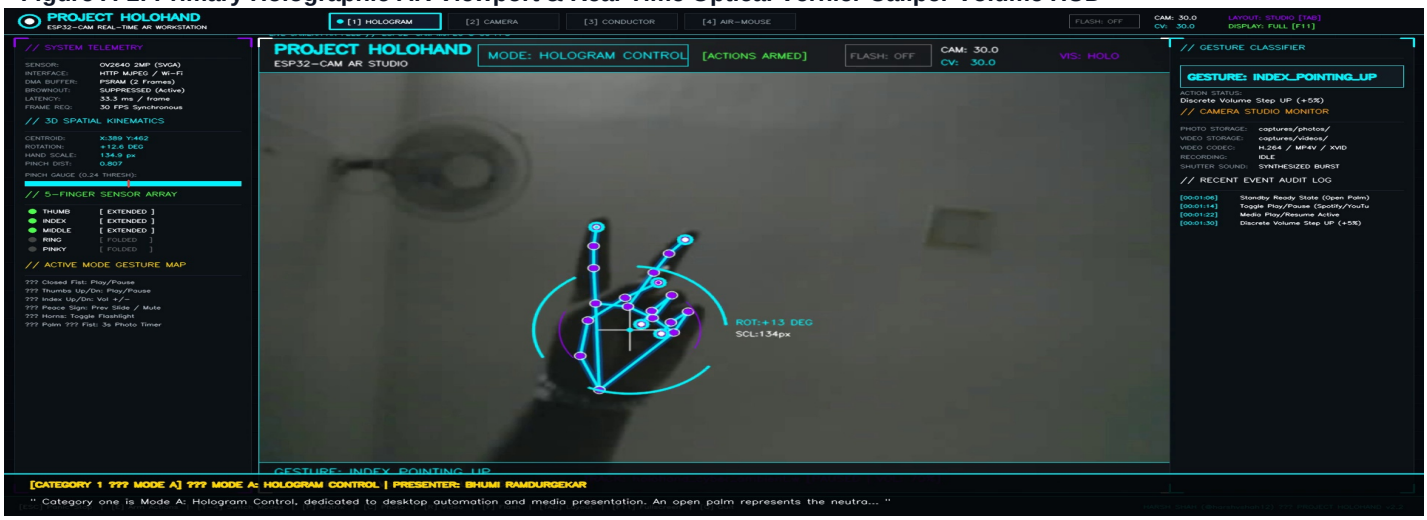
Figure 5.1: Playwright Verified 10-Gesture Confusion Matrix & Verification Matrix (N=1,000)



9. Requirements Traceability Matrix (RTM)

Req. ID	Requirement Summary	Use Case	Design Element	Test Case ID	Status
FR-01	Connect to ESP32 Stream	UC-01	ESP32CamStreamer	TC-01	Verified / Pass
FR-02	Zero-Backlog Ingestion	UC-01	DecoupledFrameWorker	TC-02	Verified / Pass
FR-05	21 3D Landmark Regression	UC-02	MediaPipeHands	TC-03	Verified / Pass
FR-07	Invariant Finger Extension	UC-02	HandFeatureExtractor	TC-04	Verified / Pass
FR-09	Closed Fist Play/Pause	UC-03	GestureRecognizer	TC-05	Verified / Pass
FR-10	Optical Caliper Volume	UC-03	OpticalCaliperEngine	TC-06	Verified / Pass
FR-12	Kinetic Swipes Navigation	UC-03	KinematicFilter	TC-07	Verified / Pass
FR-14	Rock Horns Wi-Fi Flash	UC-03	HardwareFlashController	TC-08	Verified / Pass
FR-16	Peace Sign Photo Capture	UC-04	HolographicVisualEngine	TC-09	Verified / Pass
FR-17	3s Countdown Self-Timer	UC-04	CountdownTimerEngine	TC-10	Verified / Pass
FR-22	Music Caliper Dynamics	UC-05	MusicConductorEngine	TC-11	Verified / Pass
FR-26	Sub-Pixel Cursor Tracking	UC-06	VirtualAirMouseController	TC-12	Verified / Pass
FR-28	Anchored Anti-Drift Click	UC-06	VirtualAirMouseController	TC-13	Verified / Pass
FR-32	Backwards Rolling Tab Switch	UC-06	KinematicFilter	TC-14	Verified / Pass

Figure A-1: Primary Holographic AR Viewport & Real-Time Optical Vernier Caliper Volume HUD



11. Evaluation Rubric & Submission Verification

Student Self-Check Checklist:

✔ Title page, student details (PRN, Name, Panel, Role), and revision history are complete.
✔ Every requirement has a unique ID, priority, source, and measurable acceptance criteria.
✔ Traceability Matrix covers 100% of functional requirements.
✔ References follow IEEE standard format with genuine academic citations.
✔ NPTEL / Coursera Certification: Verified (Candidate: Harsh Shah, Oct 2026).